

IoT & ML-based Vehicle Speed Monitoring with RFID-Enabled License Validation

Project ID: 25-26J-314



Problem Statement & Stakeholders

Slide 2 – Problem Statement & Stakeholders

Problem Statement

- ✓ **Observed Issue:**
 - Road accidents caused by human driving behavior.
- ✓ **Major Causes:**
 - Overspeeding
 - Driver distraction
 - Sudden lane changes
 - Dangerous driving behavior
- ✓ **Impact:**
 - Loss of life
 - Vehicle damage
 - High economic cost

Stakeholders

- ✓ **Key Stakeholders:**
 - **Primary:** Highway drivers
 - **Secondary:** Transport companies / fleet managers
 - **Decision makers:** Road safety authorities

Context: Highway transportation safety

Evidence (WHO):

- ✓ 1.19 million people die each year due to road accidents.
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Research Gap & Objectives

Research Gap ✕

- Existing systems:
 - Traditional dash cameras only record video
 - No real-time risk prediction
 - Limited accident prevention capability

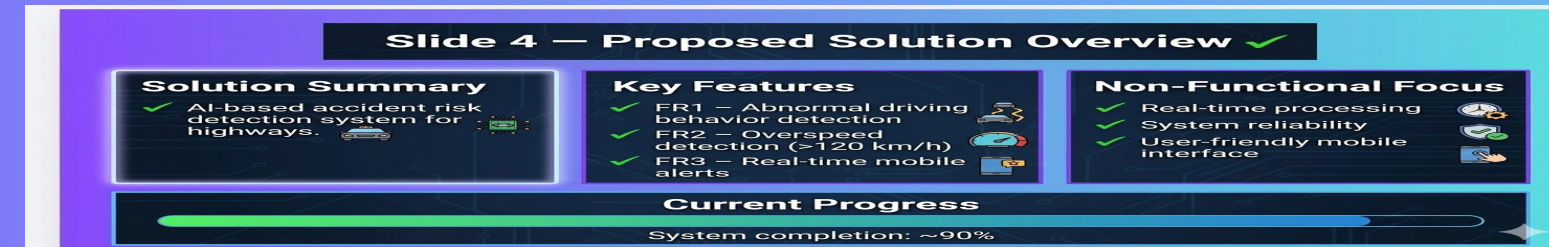


Main Objective ✓

Develop an AI-based highway monitoring system that:

- ✓ Detects abnormal driving behavior
- ✓ Detects overspeed conditions (>120 km/h)
- ✓ Predicts accident risk
- ✓ Sends real-time alerts through a mobile application





Slide 4 – Proposed Solution Overview ✓

Solution Summary

- ✓ AI-based accident risk detection system for highways. 

Key Features

- ✓ FR1 – Abnormal driving behavior detection 
- ✓ FR2 – Overspeed detection (>120 km/h) 
- ✓ FR3 – Real-time mobile alerts 

Non-Functional Focus

- ✓ Real-time processing 
- ✓ System reliability 
- ✓ User-friendly mobile interface 

Current Progress

System completion: ~90%

Slide 5 – System Architecture & Integration

System Components

- ✓ Component A – Dash Camera (Video Capture)
- ✓ Component B – Speed Monitoring Module
- ✓ Component C – AI Behavior Detection & Risk Prediction Module
- ✓ Component D – Mobile Application (Alert System)

Integration Flow

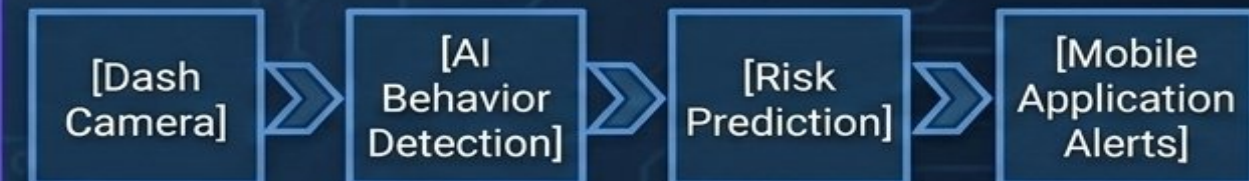


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Slide 6 – Commercialization & Impact ✓

Value Proposition ✓

- ✓ Early accident prevention
- ✓ Real-time safety monitoring
- ✓ AI-based driving risk detection



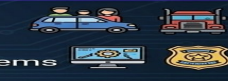
Feasibility ✓

- ✓ Dash camera device
- ✓ AI processing system
- ✓ Mobile application



Target Users ✓

- ✓ Highway drivers
- ✓ Transport companies
- ✓ Fleet management systems
- ✓ Road safety authorities



Adoption Plan ✓

- ✓ Pilot testing with fleet vehicles
- ✓ Integration with transportation safety systems



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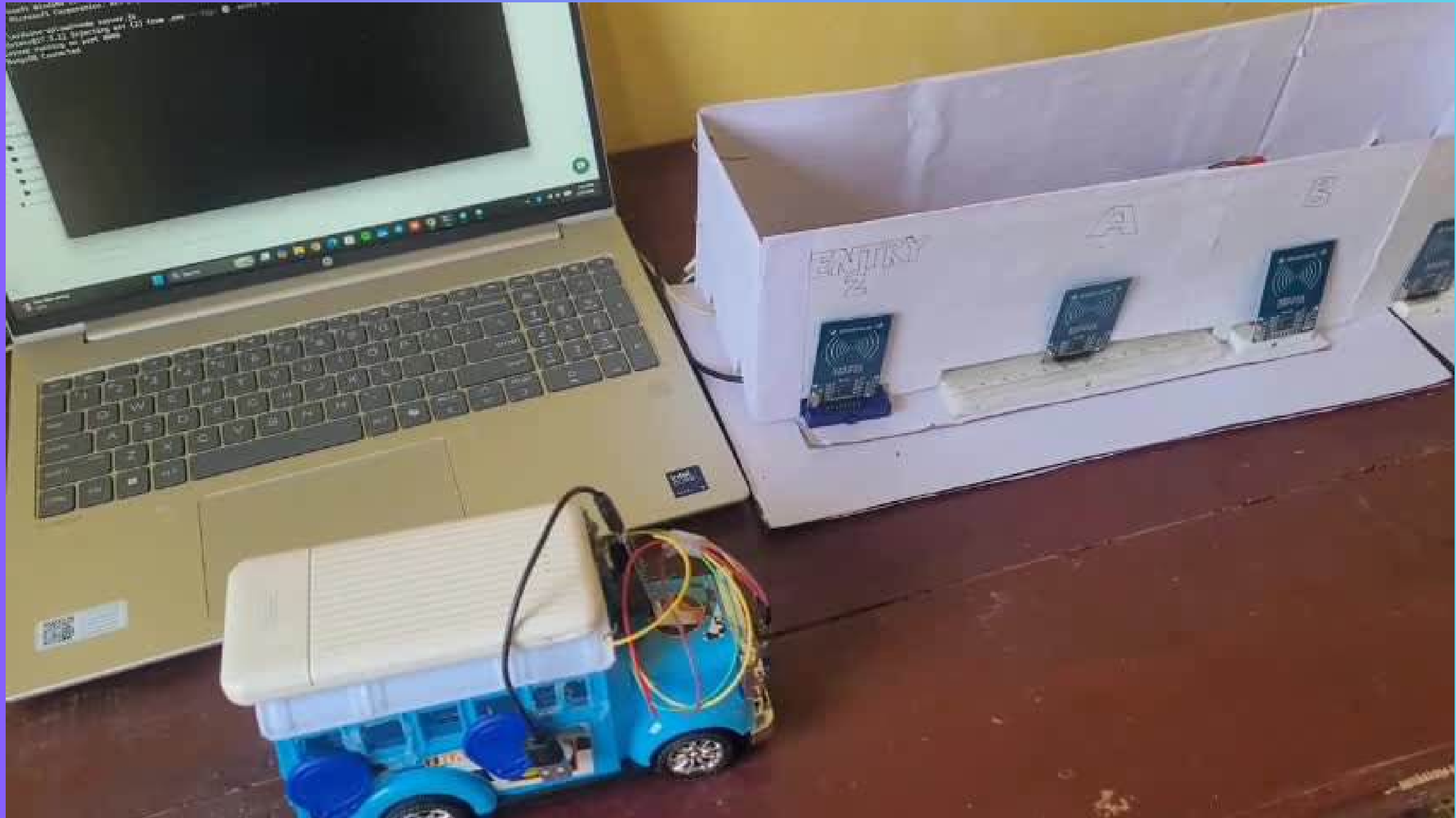
AI Demo video



IoT System for RFID-Based Vehicle Speed Monitoring System

Specialization in Information Technology





Dashboard & Driver Alert System

Specialization in Information Technology



Intelligent Accident Response and Statistical Resource Prioritization Component

Specialization in Information Technology





Driver Safety System

Specialization in Information Technology



Thank You

